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UNIVERSITY OF APPLIED SCIENCES

MBA - 2nd SEMESTER. [MBA - 10e, MIM].

Studiengang / Fachsemester Study program / Subject semester

ANALYTICAL SKILLS FOR BUSINESS.

Prüfungsfach / Prüfungsdozent:in Examination subject / Examiner

QUESTION (5).

(a) result $\leftarrow 3 + \sqrt{9} - 2^{\{2\}} * 4/3$

(b) getwd()

(c) ?? mean() ~~copy~~(d) Sales - data $\leftarrow$ "Region" = c("North", "South", "East") |>
"Sales" = c(45000, 38000, 52000).

3

2

2

missing data frame 1)

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QUESTION (1).

(a) Git - It is a version control system which is used to track, maintain & account for the changes done to a "code" or the programming language being used.

The primary purpose is to have a streamlined method of ~~tracking~~ tracking the changes made to a code, enabling multiple users to work on the base file simultaneously and check for any changes before making a "commit".

GitHub: It is the platform (majorly open-source) that offers the facilities of "Git" to its users.

The primary purpose of GitHub is to enable users to access their "Git" via an internet connection.

This allows for remote-working & interoperability of code between various users across large codebases.

Git.	GitHub.
① A version control system	A platform hosting the version control system.
② Standalone and enable access PC's by manual sharing.	Online platform & usable with an internet connection anywhere in the world.
③ Helps to track, maintain & account for the changes or updates to the codebase locally.	Helps to do all the same things but not just locally i.e. a central "hub" that can act as a check & validation for all the users across the codebase.
④ Basic support provided which is built into the language used.	Connects & allows for the user to connect with peers for advanced support.

⑥ REPOSITORY - A LOCATION WHERE THE "CODE" OR THE ^{EDITABLE} FILE WITH "THE CODE" IS STORED.
ON GITHUB - ACCESSIBLE VIA AN INTERNET CONNECTION.

COMMIT - THE ACT WHICH ALLOWS TO "PUBLISH" OR MAKE VISIBLE TO OTHERS THE CHANGES MADE TO THE CODE.

IN OTHER WORDS - COMMIT MEANS TO TELL ~~THE~~ THE PLATFORM TO PROCEED WITH MERGING THE SMALL FILE TO THE MAIN FILE.

BRANCH - BRANCH IS A SUB-CATEGORY OF THE MAIN CODE BASE WHERE UNLIMITED

NO. OF CHANGES CAN BE MADE TO THE CODE AS PER THE REQUIREMENTS. WITHOUT AFFECTING THE MAIN CODE.

AFTER WORKING ON THE SUBBRANCH, IT CAN BE PUBLISHED OR MERGED WITH THE MAIN ~~BRANCH~~ FILE OR BRANCH.
[i.e. COMMIT]

THEIR IMPORTANCE IN COLLABORATIVE PROJECTS IS SIGNIFICANT DUE TO THE FOLLOWING:

REPOSITORY - ENABLES AN ONLINE, ALLTIME AVAILABILITY OF A STORAGE LOCATION FOR ALL STAKEHOLDERS INVOLVED WHICH AVOIDS DUPLICATES OR OUTDATED FILES.

COMPLETELY REMOVES THE NEED FOR LOCAL OR OFFLINE STORAGE DEVICES FOR THE MAINTAINENCE AND INTEROPERABILITY OF THE CODE BASE.

COMMIT - ENABLES ALL USERS TO CHECK AND VALIDATE THEIR CONTRIBUTIONS TO THE MAIN CODE BASE BASED ON THE "COMMIT'S" DONE BY THE ~~PEERS~~ OTHER MEMBERS OF ~~THE~~ THE TEAMS WORKING ON COLLABORATIVE PROJECTS IN BUSINESS ANALYTICS.

ALSO ENSURES ACCOUNTABILITY AS CHANGES OR ERRORS CAN BE TRACED BACK TO THE RELEVANT USER.

BRANCH - WORKING AND ALLOWS FOR A MERGING ~~IN~~ IN TWO WAYS, ~~MERGING~~ MERGING OR DEBASING.

MERGING PRESERVES ORIGINALITY OF THE BASE ALONG WITH THE DOCUMENTATION OF THE EDITS.

DEBASING, REMOVES THE MARKERS OF EACH SUB-BRANCH AND MERGES IT LINEARLY I.E. WE LOSE ACCESS TO THE HISTORY OF EDITS.

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2(a).

SQL

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Python.

① BASIC PROGRAMMING LANGUAGE

ADVANCED PROGRAMMING
LANGUAGE WITH BASIC
FEATURES AS WELL.

② CANNOT BE USED FOR
OBJECT-ORIENTED
PROGRAMMING.

CAN BE USED FOR
OBJECT-ORIENTED
PROGRAMMING.

③ DOES NOT OFFER MANY
LIBRARIES AND ONLY
OFFERS ^{BASIC} INBUILT MATHEMATICAL
FUNCTIONS.

OFFERS GOOD AND USEFUL
LIBRARIES SUCH AS: PANDAS,
NumPy & Matplotlib

④ Useful for basic visual
representation like tables etc.

ALLOWS FOR ADVANCED
VISUAL REPRESENTATION.

⑤ DOES NOT OFFER IN-BUILT
API SUPPORT OR INTEGRATION
~~FOR~~ SUPPORT FOR ~~ADVANCED~~
ADVANCED APPLICATIONS.

OFFERS ROBUST AND
HIGH COMPATIBILITY FOR
API SUPPORT AND
INTEGRATION FOR ADVANCED
APPLICATIONS

⑥ UNRELIABLE OR LIMITED
IN THE ABILITY TO
HANDLE LARGE & SCALABLE
DATA.

SUITED TO HANDLE
MOST ~~ALL~~ TYPES OF LARGE
AND SCALABLE DATA
ESPECIALLY VISUALLY.

Database.

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2 (b). SYNTAX - THE STRUCTURE OF ANY INSTRUCTION USED TO "CODE". THE STRUCTURE INCLUDES ~~FROM~~ THE CORRECT AND MOST EFFICIENT WAY TO WRITE THE CODE.

(2b). SYNTAX - IT IS THE CORRECTLY STRUCTURED FORMAT FOR WRITING THE CODE IN ANY PROGRAMMING LANGUAGE.



GENERALLY INCLUDES OPERATORS TO START AND END ANY PROGRAMMING LANGUAGE SEQUENCE.

LIBRARIES - IT IS A SET OF FUNCTIONS WHICH ARE READILY ACCESSIBLE ~~BY~~ BY THE USER TO SAVE TIME IN WRITING MATHEMATICAL OR OTHER FUNCTIONS.



THEY ARE ALSO CALLED AS PACKAGES. AND THEY ARE MAINLY USED AS A SET OF READILY AVAILABLE COMPILATION OF NECESSARY FUNCTIONS.

EXAMPLES: tidyverse in R.

VARIABLES - THE ENTITY WHICH STORES VALUES IN THE PROGRAMMING LANGUAGE.

✓
EXAMPLE: $\boxed{X = 5} \rightarrow \text{value.}$
Variable

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FUNCTIONS - THE PRE-DEFINED SET OF INSTRUCTIONS THAT ENSURE THE PROGRAMMING LANGUAGE WORKS AS PER REQUIREMENTS.

EXAMPLE: install.packages("tidyverse")
function.

THIS function "install.packages" tells the programming language R to install the package or load the package called "tidyverse".

OBJECTS: THE ~~PRE~~ DEFINED OR CREATED SET OF "CLASSES" THAT ALLOW A USER TO DEFINE ATTRIBUTES OR PROPERTIES ~~OF~~ ~~THESE~~ ~~FUNCTIONS~~ ~~OR~~ WITHIN THE PROGRAMMING ENVIRONMENT.

CONDITIONS: THE SET OF RULES OR PARAMETERS THAT ARE VALIDATED WHEN THE CODE IS RUN OR EXECUTED.

CONDITIONS ARE THE DEFINING LOGICAL INPUT ENTERED OR CODED INTO ANY PROGRAM.

FOR SUCCESSFUL EXECUTION, THE PROGRAM ALWAYS CHECKS FOR THE CONDITIONS CODED INTO IT. BASED ON THE RESULT OF THE CHECKS, THE PROGRAM EXECUTES FURTHER.

LOOPS - THE SET OF INSTRUCTIONS THAT ARE REPETITIVE ~~AND~~ ~~AGAIN~~ ~~OR~~ ~~NOT~~ ~~NOT~~ ~~EVERY~~ ~~PART~~ OF AND CHECK AGAIN AND AGAIN TO ENSURE THE FLOW OF CODE IS SMOOTH AND ERROR-FREE.

An example: 'for', 'if' and 'while' are loops used in basic Java.

→ could not

(C) Docker² allows to export the code as an image file. The image file is light, exportable to different formats & can be used to run standalone code checks for validation.

Also allows a platform to share the selected image files online i.e. makes it cloud-based if needed. 4/1

Since it is image-based, when it is shared it can be viewed as intended by the original developer.

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② 2 popular IDE - R Studio & ~~Python~~ Power BI.

Programming languages need an "environment" to work properly & to access all their features.

This is because they need to properly load the necessary packages & define the libris. to work smoothly & properly & as intended. Also, the IDE standardises the requirement to execute & run the programming language.

The programming language is just the code, the IDE [Integrated development environment] ~~is~~ ~~the~~ is necessary to virtually represent the "code" from the programming language.

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③ 3. No-code tool - Tableau.

Lowcode tools - QGIS & Power BI.

Tableau is a open-source ~~platform~~ platform that can be used to profile →

Validate → clean → verify → document
~~validate~~ sets of data for parsing & using
for advanced programming languages &
BI [?] Software.

QGIS = ~~Quantum~~ It is a ~~Geog~~ Geographical
Information software which allows to
plot & map Geographical data all with
Contour Levels, coordinates, topography etc.

Power BI - Power Business Intelligence.
Allows to display ^{important} business analytics data
in visual format + Easy to use even
for people who ~~are~~ are not aligned
~~to~~ ~~are~~ ~~not~~ technologically ~~about~~
programming languages.

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THESE TOOLS HELP TO GENERATE CRITICAL
BUSINESS INSIGHTS WITH LITTLE TO NO
KNOWLEDGE OF PROGRAMMING. THEY ARE
SUFFICIENT FOR PENDING OR INTERNAL
VERIFICATIONS.

THEY SAVE TIME FOR HIGH LEVEL MANAGERS
AND CEOs BY SIMPLIFYING THE DECISION
MAKING PROCESS BY MAKING IT VISUAL.

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④a Mean - Gives the mean or average value of the data [From Q1 to Q4).

Median - Gives the 'mid' of the data.

Mode -

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⑥ Univariate - Analysis from one point of data.

Bivariate - Analysis from two points of data.

Multivariate - Analysis from multiple points of data.

CEOTING SALES COMPANY.

Univariate - Check for price DATA FOR REVENUE

Bivariate - Check for price & colour relation for revenue.

Multivariate - Check for price, colour, season, size for revenue

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4c. MISSING VALUES. — ~~Identify~~ Identify the reason for missing values, ignore ~~missing~~ missing without any reason, 2

Inconsistent data — check for extremes & ignore $Q1$ & $Q3$ or outlier in $Q1$ & $Q3$

Duplicate records = ~~Drop~~ De-duplication to remove exact matches.

Outliers — Ignore extreme outliers in $Q1$ & $Q3$

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